

TSU, Japan

WMO: 476510

Lat: 34.7342N

Lon: 136.5197E

Elev: 60

StdP: 14.66

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	32.0	33.5	15.4	12.0	40.5	18.4	13.9	40.8	27.7	43.5	25.6	42.6	8.9	280	0.455

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 8	(b) 10.8	(c) 92.8	(d) 77.4	(e) 90.6	(f) 77.6	(g) 88.5	(h) 77.0	(i) 80.1	(j) 88.3	(k) 79.2	(l) 87.0	(m) 78.3	(n) 85.7	(o) 10.7	(p) 310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
77.9	145.3	85.2	76.9	140.6	84.1	76.1	136.6	83.4	43.8	88.4	42.7	87.2	41.9	86.0	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 25.7	(b) 22.3	(c) 19.7	(e) 28.9	(f) 97.8	(g) 1.9	(h) 1.9	(i) 27.5	(j) 99.2	(k) 26.4	(l) 100.4	(m) 25.3	(n) 101.5	(o) 23.9	(p) 102.9		
			DB													
			WB													
				25.3	82.0	2.0	1.1	23.9	82.7	22.7	83.3	21.6	83.9	20.1	84.7	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	61.8	42.2	42.9	48.3	57.7	66.6	73.1	81.0	83.0	76.4	66.5	55.8	46.7
	DBStd	14.84	3.81	4.35	4.96	5.44	4.20	4.13	4.34	3.53	4.63	4.77	5.02	4.62
	HDD50	675	242	203	93	7	0	0	0	0	0	0	7	123
	HDD65	2978	706	618	516	224	31	0	0	0	0	39	277	567
	CDD50	4981	1	5	42	239	513	693	959	1023	793	510	182	21
	CDD65	1810	0	0	0	6	80	244	494	558	343	84	1	0
	CDH74	15703	0	0	1	23	248	1197	5045	6447	2526	216	0	0
	CDH80	5313	0	0	0	2	32	237	1840	2595	592	15	0	0
(14) Wind	WSAvg	8.6	9.6	9.5	9.4	9.0	8.8	8.0	7.7	8.3	8.4	7.6	7.6	8.8
(15) Precipitation	PrecAvg	66.7	2.1	2.6	4.4	5.3	6.9	8.2	6.9	5.9	11.5	6.7	4.0	2.1
	PrecMax	90.5	6.1	5.9	8.7	9.3	17.0	12.5	13.4	17.0	22.8	20.9	11.2	4.7
	PrecMin	39.8	0.1	0.9	1.6	1.7	2.6	3.3	1.9	0.7	1.9	1.1	0.4	0.1
	PrecStd	11.8	1.6	1.3	1.7	2.4	3.3	2.5	3.5	3.9	5.4	3.9	2.7	1.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.4	59.6	66.6	77.2	84.2	90.0	96.5	97.0	91.6	81.9	71.5	62.1
		MCWB	46.5	50.2	53.7	59.2	63.6	74.5	76.6	77.1	75.8	69.1	59.7	53.6
	2%	DB	53.4	55.2	62.2	72.0	79.4	85.7	92.8	93.4	87.8	78.8	68.4	59.2
		MCWB	45.1	47.2	51.9	58.4	64.1	71.9	77.4	77.6	75.1	67.4	59.4	50.5
	5%	DB	51.1	52.8	59.4	69.0	76.5	82.4	90.4	91.1	85.5	76.3	66.1	56.8
		MCWB	43.6	45.5	50.8	58.5	63.4	71.2	77.6	78.1	74.5	66.6	58.0	49.2
	10%	DB	49.0	50.7	57.0	66.4	74.0	80.0	88.1	89.2	83.3	74.1	64.1	54.6
		MCWB	41.9	43.9	49.3	57.3	63.1	70.3	77.0	77.6	73.8	65.1	56.6	47.4
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.6	53.5	58.4	67.2	71.2	77.0	81.2	81.8	79.3	73.8	64.3	57.1
		MCDB	53.4	56.2	61.8	70.7	76.1	85.5	89.9	89.9	85.6	78.1	66.9	59.3
	2%	WB	46.8	49.6	55.4	63.2	68.8	75.4	79.9	80.6	78.0	71.1	62.0	52.7
		MCDB	51.3	53.4	59.3	67.9	73.9	82.4	88.4	88.9	83.9	75.9	65.6	57.1
	5%	WB	44.5	47.2	53.0	61.0	67.1	74.0	78.8	79.7	76.9	68.6	60.2	50.3
		MCDB	49.8	51.4	57.4	65.7	72.6	80.1	87.3	87.6	82.6	74.0	64.3	55.4
	10%	WB	42.6	44.6	50.6	59.1	65.6	72.4	77.9	78.8	75.9	66.6	58.1	48.1
		MCDB	48.2	49.8	55.8	64.2	71.1	78.1	85.6	86.4	81.6	72.3	63.2	53.8
(35) Mean Daily Temperature Range	5% DB	MDBR	11.4	12.2	13.5	13.8	12.5	10.5	10.3	10.8	10.6	11.7	12.4	11.9
		MCDBR	14.1	14.8	16.7	17.3	17.0	14.3	14.0	13.3	12.8	13.4	13.5	14.3
	5% WB	MCWBR	9.1	9.9	10.8	9.7	8.9	6.6	5.9	5.8	6.1	7.1	8.5	9.3
		MCWBR	13.0	12.5	13.6	12.5	12.3	11.0	11.8	11.2	10.0	10.4	10.7	12.1
(40) Clear-Sky Solar Irradiance	taub		0.399	0.440	0.501	0.543	0.550	0.571	0.567	0.535	0.498	0.464	0.420	0.395
		taud	2.163	2.063	1.918	1.867	1.917	1.938	1.997	2.072	2.146	2.158	2.188	2.187
	Ebn at Noon		246	249	244	241	241	235	235	240	243	239	237	239
		Edh at Noon	37	46	58	63	61	60	56	51	45	41	36	35
(44) All-Sky Solar Radiation	RadAvg		773	987	1285	1523	1638	1463	1560	1606	1263	1016	818	700
	RadStd		51	77	87	152	180	139	210	140	116	108	76	47

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page